



Selfish-rational non-vaccination: Experimental evidence from an interactive vaccination game[☆]



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ABSTRACT

This paper provides an experimental game model – the Interactive Vaccination (I-Vax) Game – in order to investigate the behavioral consequences of risks from disease and from vaccination, and the epidemiological interdependence of vaccination decisions. Results from a controlled laboratory experiment provide evidence for selfish-rational non-vaccination: individuals react to the interactive incentive structure and make strategic vaccination decisions. We also find support for additional psychological factors determining behavior: individuals with stronger positive other-regarding preferences are more likely to vaccinate. Moreover, costs from action (vaccine-adverse events) have a stronger impact on behavior than costs from inaction (disease), which is evidence for the omission bias. Overall, we suggest that variants of the I-Vax Game can contribute to a better understanding of vaccination behavior and vaccine hesitancy. It can further be a useful experimental tool for testing interventions aiming at increasing vaccine uptake.

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1. Introduction

Vaccines against infectious diseases are among the most effective preventive methods available to modern healthcare, saving approximately three million human lives per year (WHO, 2012). Nevertheless, vaccine hesitancy is a severe threat to the success of national and international immunization programs, which are of utmost importance for the prevention and control of morbidity and mortality within societies. In fact, vaccination rates repeatedly fall below the critical thresholds which are necessary to eliminate diseases such as measles or polio. This drop in vaccination rates repeatedly jeopardizes the attainment of global public health goals (WHO, 2013). It is therefore an important challenge for the medical and the social sciences to understand people's underlying motivation to decide in favor of or against vaccination. Understanding the processes behind vaccine hesitancy will enable the creation of more effective interventions in order to finally achieve public health goals (Betsch et al., 2015).

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Vaccination has been typically described as an individual decision making task in which the risks of being unvaccinated and contracting the disease are weighted against the risks of getting vaccinated, which may lead to mostly mild – but in some very rare cases also to severe – adverse events. As the risks involved in getting vaccinated are usually lower than the risks involved with the disease itself, vaccination constitutes a rational choice (Weinstein, 2000). Thus, vaccine hesitancy can be seen as a result of biased information processing or ill-informed decision making, for example because the risks involved in getting vaccinated are skewed and inflated (e.g., Betsch et al., 2011; Brewer et al., 2007; Chen, 1999; Hollmeyer et al., 2009; WHO, 2013). For instance, it has been shown that the same symptom occurring as a consequence of vaccination is evaluated more negatively than when it is evaluated as a symptom during the corresponding illness (Brown et al., 2010). This has been interpreted to be the result of the omission bias, that is, the tendency to evaluate negative events as more negative when they are due to an action rather than due to an inaction (e.g., Spranca et al., 1991; Asch et al., 1994).

The individual perspective focuses only on the direct effect of vaccination, i.e., the probability of certain consequences as a result of a person's own vaccination decision. According to this perspective, non-vaccination must be the results of psychological biases or errors. However, vaccination also yields an indirect effect because it reduces the transmission of a disease (Anderson and May, 1985). Considering this indirect effect, non-vaccination might not be the result of biased decision making but rather the result of selfish-rational behavior. Accordingly, the decision in favor of vaccination not only benefits the decision maker but also provides a positive externality on other individuals in the population because it decreases the likelihood that the disease spreads, an effect called *herd immunity* (Fine et al., 2011; Stiglitz, 2000). This implies: the more individuals are vaccinated in a population, the less attractive (relative to non-vaccination) vaccination is because an infection becomes less likely, while the likelihood of vaccine-adverse events remains stable. The individually selfish-rational solution (non-vaccination) may therefore differ from the collectively optimal solution (vaccination), leading to a social dilemma (Dawes, 1980; Kollock, 1998).

A growing body of theoretical and simulation-based evidence demonstrates that strategic-interactive reasoning and other-regarding preferences may be relevant for real-world vaccination behavior, that is, that individuals may free-ride out of rational selfishness or vaccinate out of a prosocial motivation (e.g., Bauch and Bhattacharyya, 2012; Bauch and Earn, 2004; Bauch et al., 2003; Cohen et al., 2013; Galvani et al., 2007; Manfredi et al., 2009; Oraby et al., 2014; Madrian, 2014). As a prominent example, Bauch and Earn (2004) use game-theoretic models to show that due to the indirect protection provided by herd immunity, even small increases in perceived vaccine risk may lead to declines in vaccine uptake. They conclude that “[...] it is impossible to eradicate a disease through voluntary vaccination when individuals act according to their own interests” (p. 13392). Furthermore, a review of surveys and interview studies by Quadri-Sheriff et al. (2012) suggests that the indirect effect of herd immunity is indeed considered by parents when deciding whether or not to immunize their children. However, there is only little behavioral evidence as to whether people do indeed react to the interactive and dynamic incentive structure of vaccination decisions. It is important to note that previous work using experimental games neglect the psychological consequences of vaccine-adverse events.

The present paper provides an experimental game model that integrates the epidemiological and economic interdependence of individuals' vaccination decisions (e.g., Bauch and Earn, 2004; Fine et al., 2011). With this, we are able to investigate the psychological consequences of risks from disease and from vaccination, and their effects on vaccination behavior in a realistic interaction setting. In a controlled laboratory experiment, we test for selfish-rational non-vaccination (i.e., free-riding on others' indirect protection). Furthermore, we investigate additional psychological factors that may either increase (i.e., positive other-regarding preferences) or decrease (i.e., omission bias) the likelihood of non-vaccination. We show that individuals react to the interactive incentive structure and get vaccinated strategically. Prosocial individuals, i.e. those who also regard the outcomes of others in their decisions, are more likely to vaccinate than prosocial individuals, who focus solely on their own outcome. Additionally, participants are more likely to change their behavior if they encounter negative consequences after vaccination (vaccine-adverse events) rather than after non-vaccination (disease). Thus, we find support for selfish-rational behavior with regard to vaccine uptake and vaccine hesitancy as well as additional psychological factors determining vaccination behavior. Our experimental game model helps to better understand the motivations underlying vaccine hesitancy and vaccination by adding evidence that also rational calculation can be a reason for non-vaccination (cf. Betsch et al., 2015). It can serve as a tool to pilot interventions in order to finally increase vaccination uptake.

The remainder of the paper proceeds as follows: In Section 2, we discuss related experimental literature on vaccination as a strategic interaction, Section 3 devises the novel game, Section 4 presents the methods and results of a laboratory experiment, and Section 5 discusses the findings and provides conclusions.

2. Related literature

There have been only a few attempts to experimentally explore vaccination behaviors or intentions, while considering both the direct and indirect effects of vaccination. In an experimental survey study, Betsch et al. (2013) found that vaccination intention decreased when the individual benefit from herd immunity was made salient, whereas vaccination intention increased when other people's benefit was emphasized (i.e., social benefit), given that the vaccination costs were low. In a similar vein, Vietri et al. (2012) presented participants with hypothetical vignettes, orthogonally varying the individual's own risk of infection and the number of people who would indirectly benefit from that individual's vaccination. They particularly showed that under conditions of low risk of infection, vaccination intentions increased, the more others benefited from the vaccination. Moreover, Attari et al. (2014) found that the second most important reason for an individual to have the

influenza vaccination is to prevent the disease from spreading. Although these findings point in a direction where both self-interest as well as other-regarding preferences (e.g., altruistic or social welfare concerns) might affect vaccination behavior, these studies did not investigate real interactions and behavior. Furthermore, the measured behavioral outcomes were hypothetical and not incentivized.

Chapman et al. (2012) and Ibuka et al. (2014) used interactive laboratory experiments to model the incentives from direct and indirect effects of vaccinations. In a multi-round game with groups of 8–10, players were endowed with valuable points. In each round, they decided either in favor of or against a costly vaccination. Rejecting vaccination yielded the risk of costly infection. It was found that players react to other group members' vaccination decisions (i.e., vaccination decreased, the more other players have got vaccinated in the previous round; Ibuka et al., 2014) and that vaccination was more likely when its expected value increased relative to non-vaccination (Chapman et al., 2012). Neither study took potential vaccine-adverse events into account. The occurrence and perception of vaccine-adverse events, however, is an important determinant of vaccination decisions, because it has implications for the incentive structure, as will be explicated below. Therefore, we propose a new experimental game of vaccination decisions – the Interactive Vaccination (I-Vax) Game – which is both epidemiologically and game-theoretically well-grounded, and considers the risks of both vaccination and non-vaccination.

3. The Interactive Vaccination (I-Vax) Game

The game is played by N players i over r rounds. Each player is endowed with e experimental currency units (ECUs) in each round. In each round, players have to choose exactly one behavioral action $a_i \in \{0, 1\}$, representing *non-vaccination* and *vaccination*, respectively. The expected utility (EU_i) is expressed by the following function:

$$EU_i = e - c_{a_i}^{\text{fix}} - c_{a_i}^{\text{var}}$$

Vaccination yields fixed costs (c_1^{fix} , e.g., time, money), whereas non-vaccination is costless ($e > c_1^{\text{fix}} > c_0^{\text{fix}} = 0$). Furthermore, variable costs ($c_{a_i}^{\text{var}}$) in the case of both vaccination and non-vaccination may occur, represented by the following function:

$$c_{a_i}^{\text{var}} = p_{a_i} \times s_{a_i,j}$$

There is a certain probability (p_{a_i}) that both vaccination and non-vaccination will have negative consequences ($s_{a_i,j}$) due to vaccine-adverse events or infection with the disease, respectively. This probability is fixed in the case of vaccination (p_1), whereas it is uncertain in the case of non-vaccination. In the latter case, the probability of infection varies as a function of the basic reproduction number of the infectious disease (R_0 , with $R_0 > 1$), and the number of vaccinated players in the current round (based on a homogeneously mixed SIR model with birth and death). The basic reproduction number (R_0) indicates how contagious a disease is by representing the number of individuals that one infected person will infect, given that no individual in the population has been vaccinated. Moreover, the more people who are vaccinated, the smaller the probability of contracting the disease is for unvaccinated individuals (see e.g., Fine et al., 2011):

$$p_{a_i} = \begin{cases} p_1 & \text{if } a_i = 1 \\ 1 - \frac{1}{R_0 \left(1 - \frac{\sum_{k=1}^N a_k = 1}{N}\right)} & \text{if } a_i = 0 \wedge R_0 \left(1 - \frac{\sum_{k=1}^N a_k = 1}{N}\right) \geq 1 \\ 0 & \text{if } a_i = 0 \wedge R_0 \left(1 - \frac{\sum_{k=1}^N a_k = 1}{N}\right) < 1 \end{cases}$$

Given that negative consequences after vaccination (vaccine-adverse event; $s_{1,j}$) or non-vaccination (infection; $s_{0,j}$) occur, they will be of different severity $j \in \{\text{mild}, \text{medium}, \text{severe}\}$:

$$j = \begin{cases} \text{mild} & \text{with probability } p_{\text{mild}} \\ \text{medium} & \text{with probability } p_{\text{medium}} \\ \text{severe} & \text{with probability } p_{\text{severe}} \end{cases}$$

On the one hand, mild consequences are more likely than consequences of medium severity, which are in turn more likely than severe consequences, $p_{\text{mild}} > p_{\text{medium}} > p_{\text{severe}}$, considering $p_{\text{mild}} + p_{\text{medium}} + p_{\text{severe}} = 1$.¹ On the other hand, mild consequences are less costly than medium consequences, which are in turn less costly than severe consequences; and the

¹ For simplicity, the distribution of the probabilities for mild, medium, or severe consequences is assumed to be equal for vaccine-adverse events and infections. The unconditional probability for each possible negative consequence is calculated by $p_{a_i} p_{\text{mild}}$, $p_{a_i} p_{\text{medium}}$, and $p_{a_i} p_{\text{severe}}$, whereas the probability for the opposite event to have no negative consequences after vaccination or non-vaccination is given by $1 - p_{a_i}$.

negative consequences that may occur after non-vaccination (i.e., symptoms of the disease) are at least as negative as the consequences that may occur after vaccination, $s_{1,mild} \leq s_{0,mild} \leq s_{1,medium} \leq s_{0,medium} \leq s_{1,severe} \leq s_{0,severe}$. Finally, if no other players have been vaccinated ($\sum_{k \neq i} a_k = 0$), a player i 's expected utility from vaccination is greater than her expected utility from non-vaccination, which can be simplified to

$$c_1^{fix} + p_1 \times s_{1,j} < \left(1 - \frac{1}{R_0}\right) s_{0,j}$$

A rational self-interested player would choose the behavioral action that maximizes her individual payoff. The expected payoff from vaccination remains constant, while the expected payoff from non-vaccination increases with an increasing fraction of vaccinated players. Non-vaccination is the unique best response if the vaccination rate exceeds a certain threshold. More precisely, vaccination is the best response if

$$c_1^{fix} + p_1 \times s_{1,j} < \left(1 - \frac{1}{R_0(1 - (\sum_{k=1}^N a_k = 1/N))}\right) s_{0,j}$$

A certain fraction of vaccinated individuals constitutes a Nash equilibrium if all players – vaccinated and non-vaccinated – cannot increase their expected payoff by another behavioral action. Social welfare (i.e., the players' aggregated expected payoffs) is maximized if the pathogen cannot spread anymore and is eliminated from the population. If this critical herd immunity threshold (e.g., [Fine et al., 2011](#)) is reached at a larger fraction of vaccinated individuals than in equilibrium, the situation poses a social dilemma.

4. Experiment

In order to investigate choice behavior in the I-Vax Game, a laboratory experiment was conducted. The goal was to establish the I-Vax Game as a valid experimental tool for studying vaccination decisions and to qualify the tool as a method for identifying factors to increase vaccine uptake. Vaccination behavior might be affected by a range of non-monetary factors, such as attitudes, opinions, and direct or indirect experiences, which are specific to the framed context. According to [Chong and Druckman \(2007\)](#), “[f]raming refers to the process by which people develop a particular conceptualization of an issue or reorient their thinking about an issue.” (p. 104). Framing strategic interactions has been shown to affect choice behavior in various domains (e.g., [Druckman, 2001](#); [Dufwenberg et al., 2011](#); [Cookson, 2000](#); [Bougherara et al., 2011](#); [Iturbe-Ormaetxe et al., 2011](#); [Rege and Telle, 2004](#)). Thus, it is necessary to study if the specific framing as a vaccination decision influences behavioral decisions. The experimental setup therefore contains a *Framing* and *No-Framing* treatment. In the *Framing* treatment, the decision was labeled as vaccination vs. non-vaccination against an infectious disease, with potential vaccine-adverse events, all causing potential losses of fitness points. In the *No-Framing* treatment, the game's incentive structure and procedure were exactly replicated, but the options and outcomes were framed in a neutral way: option A (being structurally equivalent to vaccination) or not option A (equalling non-vaccination), point loss (equivalent to disease or vaccine-adverse events), and endowment (fitness points).² In the following sections, we provide a detailed description of the method and formulate hypotheses regarding the expected behavior in the I-Vax Game.

4.1. Method

4.1.1. Participants and design

Eight experimental sessions were conducted in spring/summer 2014 at the Erfurt Laboratory for Experimental Economics (*eLab*), University of Erfurt in Germany. A one-factorial between-subjects design was applied: *Framing* versus *No-Framing*. Four sessions with two groups each were assigned to the *Framing* treatment ($N=96$, 8 independent observations). The other three sessions with two groups each and one session with one group (due to no-show ups) were assigned to the *No-Framing* treatment ($N=84$, 7 independent observations). Participants were 56 male and 124 female social science students of the University of Erfurt with an average age of $M=23.37$ years ($SD=4.09$).

4.1.2. Decisions in the experimental game

The decision in the I-Vax Game, either vaccination or non-vaccination, was the dependent variable. The I-Vax Game had the following parametrization: Participants played in groups of $N=12$ over $r=20$ rounds. Group composition remained constant over the rounds (*partner matching* protocol). Players were endowed with $e=100$ ECUs in each round. The basic reproduction number of the disease was set to $R_0=3$. This parametrization is comparable to a highly contagious influenza strain (e.g. as during the Spanish flu; see [Mills et al., 2004](#)). The repeated character of the game also resembles the annual vaccination against influenza.

On the one hand, vaccination yielded fixed costs of $c_1^{fix}=5$ and the probability of vaccine-adverse events was $p_1=0.3$, with the following point losses in the case of mild, medium, and severe consequences: $s_{1,mild}=21$, $s_{1,medium}=36$, $s_{1,severe}=51$. On the

² For the sake of simplicity, we will refer to vaccination and non-vaccination decisions, irrespective of treatment.

other hand, in the case of non-vaccination, potential consequences were set to $s_{0,mild} = 35$, $s_{0,medium} = 60$, $s_{0,severe} = 85$. For both vaccination and non-vaccination, the probability of mild, medium, and severe consequences was $p_{mild} = 0.5$, $p_{medium} = 0.4$, and $p_{severe} = 0.1$, respectively. Hence, the average negative consequence of an infection was $s_0 = 50$, whereas the average negative consequence of vaccine-adverse events was $s_1 = 30$.

After each round, participants received summary feedback about (1) their own vaccination decision, (2) the fraction of all participants vaccinated (in percent), (3) the resultant probability of infection, (4a) their points lost (if any) in the case of vaccine-adverse events after vaccination, or (4b) their points lost (if any) in the case of infection after non-vaccination, and (5) their points earned in the previous round (endowment minus points lost).

According to this parametrization, a selfish-rational player has no incentive to change her behavioral action – irrespective whether this has been vaccination or non-vaccination in the previous round – if 7 players have been vaccinated overall, whereas she has an incentive to switch from non-vaccination to vaccination if 6 players have been vaccinated ($EU_i = 83.3$ if $a_i = 0$ and $EU_i = 86$ if $a_i = 1$, respectively), and to switch from vaccination to non-vaccination if 8 players have been vaccinated ($EU_i = 90$ if $a_i = 0$ and $EU_i = 86$ if $a_i = 1$, respectively). Therefore, a share of 0.58 (i.e., 7 players) vaccinated players constitutes a Nash equilibrium. However, this would lead to a socially inefficient vaccination rate. Social welfare is maximized by a share of 0.67 (i.e., 8 players) vaccinated players, in which case the probability of infection for non-vaccinated individuals is zero and the aggregated payoffs reach their maximum (7 players vaccinated: $\sum_{k=1}^N EU_k = 1052$, 8 players vaccinated: $\sum_{k=1}^N EU_k = 1088$, 9 players vaccinated: $\sum_{k=1}^N EU_k = 1074$).³

4.1.3. Social value orientation (SVO)

As a measure of other-regarding preferences, i.e., an individual's preferences for another individual's outcome in relation to her own outcome, social value orientation (SVO) was assessed with a z-Tree implementation (Crosetto et al., 2012) of the paper-based SVO slider measure (Murphy et al., 2011). In this measure, participants make 6 independent allocation decisions of decomposed games (i.e., dictator games), distributing points as a sender between themselves and an unknown receiver (for an example item, see Appendix B-I). In these allocations, giving more points to oneself reduces the payoff of the receiver and vice versa. Furthermore, giving more points to the receiver may not only increase equality between the sender and the receiver's payoff but may also increase the sum of joint payoffs, i.e., social welfare. Hence, altruistic, equality, and/or efficiency concerns determine prosocial allocations (Van Lange, 1999). Responses can be scored to yield a single angle-index of a participant's SVO, where low values indicate a rather prosocial orientation and high values indicate a rather prosocial orientation.⁴ Hence, this score may be used to classify the person as either prosocial or prosocial. Murphy et al. (2011) demonstrated a very good test-retest reliability (89% consistency of classification) and good predictive validity.

The strategy method (Selten, 1967) was used to estimate SVO scores of all participants: the roles (sender vs. receiver) were allocated randomly at the end of the experiment. Additionally, only one dictator game allocation of the sender was randomly selected and became relevant for the sender's and respective receiver's payoff.

4.1.4. Vaccination attitude

The attitude towards vaccination is “a relatively enduring organization of beliefs, feelings, and behavioral tendencies” towards vaccination (Hogg and Vaughan, 2005, p. 150) and is accessible as the general personal evaluation of vaccinations which can range from extremely negative to extremely positive. It is assumed to predict behavioral intentions and behavior (Ajzen, 1991). Previous studies showed that the attitude towards vaccinations is an important predictor of vaccination behavior (Opel et al., 2013, Lindley et al., 2006). Hence, if the vaccination attitude predicts behavior in the I-Vax Game as well, this would be evidence for the external validity of the game model. Furthermore, very positive or negative attitudes towards vaccination may reduce strategic vaccination behavior. Therefore, participants' general attitude towards vaccination was assessed with a 100-point bipolar slider item: “In general, I am...” (1 = *absolutely against vaccinations*, to 100 = *absolutely in favor of vaccinations*).

4.1.5. Procedure

Participants were recruited via the online registration software ORSEE (Greiner, 2004) to take part in a computerized decision making experiment implemented with the software z-Tree (Fischbacher, 2007). On arrival, participants drew an index card to determine their cubicle number. The experiment had two separate parts: an individual decision making task and a group decision making task. Decisions in one randomly selected game of the individual decision making task and all rounds of the group decision making task were incentivized. All payoffs of the experiment were determined in points and

³ See Appendix A for a detailed overview of the individual expected payoffs in case of vaccination and non-vaccination and the aggregated population payoff for each fraction of individuals vaccinated.

⁴ There are various ways how to conceptualize and measure other-regarding preferences (Murphy and Ackermann, 2014). In the SVO slider measure, the angle of an individual is calculated as follows: $SVO^\circ = \arctan((\text{mean allocation other} - 50)/(\text{mean allocation self} - 50))$. Note that the value of 50 is subtracted in order to shift the center of the resulting ring to the origin of the Cartesian plane (for details see Murphy et al., 2011). The more points the sender allocates to the receiver and the less points the sender allocates to herself out of altruistic, equality, and/or efficiency concerns, the larger the SVO angle, i.e., larger angles indicate stronger positive other-regarding preferences.

converted into Euros at the end of the experiment (conversion rate: 100 points = 0.50 Euros). On average, participants earned 8.90 Euros ($SD = 0.44$) for about one hour of participation.

Participants first received printed instructions, providing general information and the rules of the individual decision making task, including an example (slider measure to assess social value orientation, see Appendix B-I). After the experimenter had read the instructions aloud, participants could ask clarifying questions. Participants then completed the individual decision making task. Afterwards, they received printed instructions on the group decision making task, including some examples (I-Vax Game, see Appendix B-II). Again, instructions were read aloud and participants could ask questions. Before the group decision making task started, all participants had to correctly answer test questions (see Appendix C). After completion of the group decision making task, participants answered some post-experimental questions including their vaccination attitude and demographics. They were informed about their individual payoff for the two parts and their overall payoff in Euro. Finally, all participants received their earnings privately.

4.1.6. Data and analysis

Participants made repeated decisions and received feedback about other group members' decisions as well as about their own outcomes after each round. Therefore, individual observations are nested within subjects, rounds, and groups. In order to consider this clustered data structure by controlling for the interrelated error terms, we applied mixed effects models when estimating predictors of vaccination decisions (with a logit link, accounting for the binary format), including random effects of the subject, round, and group (random intercepts with fixed means, independent for each variable). Regression models were estimated using the *lme4* package (Bates et al., 2014) in the R-environment (R Development Core Team, 2008) and have the following form:

$$Pr(a = 1) \sim \text{fixed effects} + (1 \mid \text{subject}) + (1 \mid \text{round}) + (1 \mid \text{group}) + \varepsilon$$

4.2. Hypotheses

Following previous experimental and non-experimental research (e.g., Bauch and Earn, 2004; Betsch et al., 2013; Chapman et al., 2012; Ibuka et al., 2014), we expect strategic adjustment of behavior contingent to feedback, which is provided after each round. In the I-Vax Game, vaccination is the best response if the fraction of other players getting vaccinated is expected to be below the Nash equilibrium, whereas non-vaccination is the best response if the opposite is true. Hence:

H1. Vaccine uptake should increase (decrease) if the fraction of other players who got vaccinated was below (above) the Nash equilibrium in the previous round.

One of our basic assumptions is that strategic considerations affect vaccination decisions. We expect that individual differences in other-regarding preferences – assessed as the individual's social value orientation (SVO; for overviews, see Murphy and Ackermann, 2014; Rusbult and Van Lange, 2003; Van Lange et al., 2007) – will determine vaccination behavior, too. Empirical evidence from various social dilemmas supports the assumption that individuals with a focus on their own welfare – *proselfs* – are less cooperative than people with positive other-regarding preferences – *prosocials* (e.g., De Cremer and Van Vugt, 1999; De Dreu, 2010; Parks, 1994). Since eradicating the disease protects all players, stronger altruistic, equality, and efficiency concerns should lead to a greater likelihood of vaccine uptake. Therefore:

H2. Vaccine uptake among prosocials is higher than among proselfs.

The attitude towards vaccination has been shown to be a determinant in vaccination decision making (Godin et al., 2010; Herzog et al., 2013; Hollmeyer et al., 2009). The general attitude towards vaccination should impact behavior only if the participant perceives the situation as a vaccination decision, that is, when the situation is framed as a vaccination decision. Hence:

H3. The attitude towards vaccination predicts vaccine uptake in the I-Vax Game in the *Framing* treatment but not in the *No-Framing* treatment.

Each individual can experience negative consequences from her decision – vaccine-adverse events in the case of vaccination and contracting the disease in the case of non-vaccination. Previous research has shown that people are more sensitive to negative consequences that occur as a result of their own actions than as a result of inactions, which constitutes the so-called omission bias (Spranca et al., 1991; Ritov and Baron, 1990; Kahneman and Tversky, 1982). Therefore:

H4. Experiencing negative consequences in the previous round changes behavior in the subsequent round. Specifically, if the behavior leads to negative consequences due to vaccine-adverse events in the case of vaccination or due to infection in the case of non-vaccination, respectively, players are more likely to change their behavior in the subsequent round than when the behavior caused no negative consequences. This effect is larger if negative consequences occur due to an action (i.e., vaccination) than due to an inaction (i.e., non-vaccination).

4.3. Results

4.3.1. Descriptives

SVO scores of participants range from -6.76° to 48.08° , with an average score of 27.74° ($SD = 12.45$). Given the angles of idealized SVO types, a classification can be derived with angle-scores of prosocials (i.e., seeking to maximize equality between own and others' outcomes, and/or social welfare) between 22.45° and 57.15° , and proselves (i.e., seeking to maximize the own payoff) ranging between -12.04° and 22.45° (see Murphy et al., 2011).⁵ Accordingly, 124 participants (69%) were classified as prosocial and 56 participants (31%) were classified as prosself.

Across conditions, participants had a positive attitude towards vaccination ($M = 71.83$, $SD = 24.42$). Nevertheless, participants substantially alternated their vaccination decision over rounds. Overall, 154 participants (86%) switched at least once from vaccination to non-vaccination or vice versa, whereas only 26 participants either never got vaccinated (7%) or always got vaccinated (7%) over all rounds.⁶ Moreover, there is a significant rank-order correlation between vaccination attitude and switching behavior in the *Framing* treatment – those with a positive attitude vaccinate more often and rather do not switch to non-vaccination while those with a neutral attitude have higher switching rates ($r_s = .350$, $p < .001$; with mean attitude $M = 40.29/68.00/88.20$ for those who never/sometimes/always get vaccinated). In contrast, there is no such relation in the *No-Framing* treatment ($r_s = -.138$, $p = .212$; with $M = 86.00/75.44/73.00$ never/sometimes/always get vaccinated).

Fig. 1 shows the average fraction of individuals getting vaccinated per round collapsed across SVO types. In the upper part of Fig. 1 it can be seen that the average number of vaccinated individuals in the *Framing* treatment ($M = 0.52$, $SD = 0.05$) and in the *No-Framing* treatment ($M = 0.50$, $SD = 0.04$) is closer to the Nash equilibrium (0.58) than to the social welfare optimum (0.67). Moreover, although there is considerable variance in vaccination decisions over rounds on the individual level (see above), the average fraction of individuals per group getting vaccinated seems to be rather stable over rounds. The lower part of Fig. 1 collapses across framing treatments and illustrates that proselves ($M = 0.45$, $SD = 0.06$) get vaccinated less often than prosocials ($M = 0.53$, $SD = 0.05$).

4.3.2. Vaccination decisions as a strategic interaction

Hypothesis 1 states that individuals change their decision strategically as a response to the number of other players who got vaccinated. In other words, we test whether non-vaccination can be the result of selfish-rational behavior. In order to test this hypothesis, a dummy variable *RelVacc* was created. This variable is coded with 1 if the relative fraction of players who got vaccinated was above the Nash equilibrium in the previous round, and is coded with 0 when it was below the threshold. Model 1 (see Table 1) includes treatment (0 = *Framing*, 1 = *No-Framing*) and *RelVacc* as predictors of the vaccination decision (0 = non-vaccination, 1 = vaccination).⁷

Supporting Hypothesis 1, *RelVacc* significantly predicts vaccine uptake: If the fraction of players who got vaccinated was above the equilibrium in the previous round, such that non-vaccination yielded a higher expected payoff than vaccination, vaccination decreased in the following round and vice versa.⁸ Although the interaction effect does not reach conventional criteria of significance ($p = .120$), the effect appears to be mainly driven by the *Framing* treatment: A simple slope analysis supports this idea. Participants in the *Framing* treatment react to the fraction of players who got vaccinated ($B = -0.321$, $SE = 0.152$, $p = .035$), whereas there is no such effect in the *No-Framing* treatment ($B = 0.078$, $SE = 0.172$, $p = .651$).

Model 2 includes participants' social value orientation classification (0 = prosocial, 1 = prosself) as an additional predictor. Whereas the effect of *RelVacc* decreases slightly but remains marginally significant, SVO is also marginally significant, yielding weak support for Hypothesis 2. Prosocials, i.e., participants who value the outcomes of others to a greater extent than proselves, were thus more likely to get vaccinated than proselves.⁹

Finally, model 3 adds the general attitude towards vaccination as a predictor. The analysis reveals significant main effects of treatment, *RelVacc*, and general vaccination attitude, all qualified by significant interaction effects. As predicted in Hypothesis 3, the general attitude towards vaccination is a stronger predictor of decisions when the game is framed as a

⁵ Resulting angles can also be outside of these ranges, yielding a classification as altruistic or competitive. However, none of our participants had such a score.

⁶ This pattern appears to be consistent in both experimental conditions, with 7% never getting vaccinated in the *Framing* and *No-Framing* treatment, 82% switching in the *Framing* treatment and 89% in the *No-Framing* treatment, and 10% always getting vaccinated in the *Framing* treatment and 4% in the *No-Framing* treatment. The switching behavior did not differ between treatments (Fisher's exact test: $p = .208$). Similarly, the general attitude towards vaccination did not differ between treatments (Wilcoxon rank-sum test: $p = .114$).

⁷ Including participants' age and gender as covariates does not qualitatively change any of the reported results. We report models including all the possible interaction terms of predictor variables. For parameter estimates of models considering only main effects, see Appendix D.

⁸ Additional analyses of predicting switching behavior also underpins the finding: the dependent variable is coded with 0, if a player switched from vaccination in the previous round to non-vaccination in the current round. In the case of switching from non-vaccination to vaccination, the dependent variable is coded with 1. The independent variable is coded with 0 if the relative fraction of players who got vaccinated in the previous round was exactly in equilibrium (i.e., 7 players who vaccinated), and is coded with 1 (2) if it was below (above) this threshold. Results reveal that switching to vaccination was significantly more likely if the fraction of vaccinated individuals was below than when it was in equilibrium ($B = 1.167$, $SE = 0.226$, $p < .001$), and switching to non-vaccination was significantly more likely when the vaccination rate was above rather than in equilibrium ($B = -0.628$, $SE = 0.281$, $p = .026$).

⁹ Building subsets of the data with different intervals of other players vaccinated in the preceding round (0 to NE and NE to SWO), we find significantly more vaccine uptake among prosocials compared to proselves in both intervals (both $ps < .1$). Hence, it appears that prosocials react similarly to the strategic incentives as proselves do, however, they have a higher intercept of vaccination.

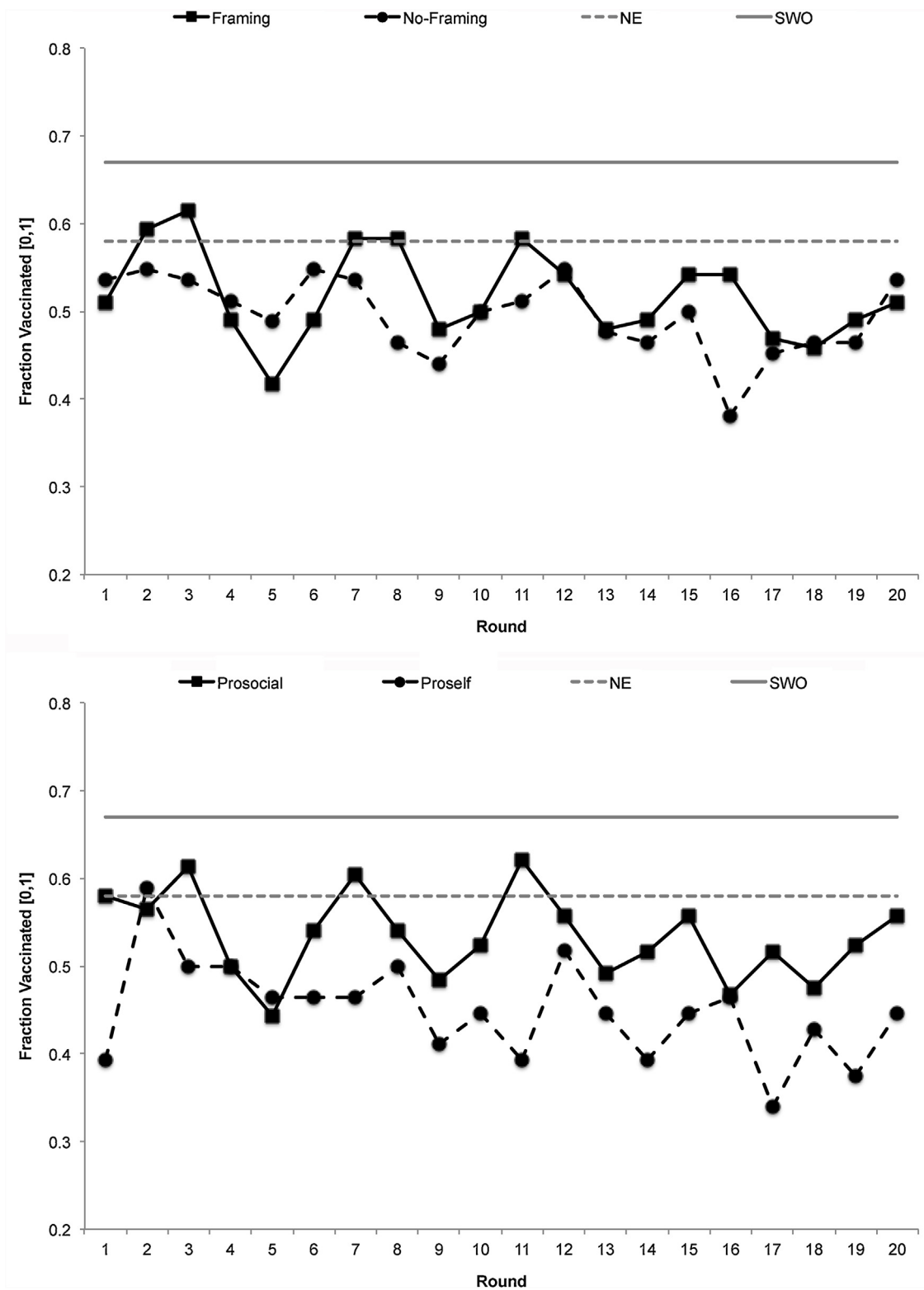


Fig. 1. Fraction vaccinated by experimental treatment (top) and social value orientation (bottom). NE = Nash equilibrium, SWO = social welfare optimum. Data is collapsed across social value orientation types (top) and framing treatments (bottom).

Table 1

Mixed effects models predicting decisions in favor of vaccination by the relative fraction of players who got vaccinated in the previous round (reference: fraction vaccinated below Nash equilibrium), experimental treatment (reference: *Framing* treatment), social value orientation (reference: prosocial), and general attitude towards vaccination.

Predictor	Model 1		Model 2		Model 3	
	B	SE	B	SE	B	SE
(Intercept)	0.280	0.203	0.508	0.244*	−2.454	0.656***
Treatment: No-Framing (A)	−0.318	0.289	−0.436	0.344	4.219	1.073***
RelVacc: above NE (B)	−0.295	0.147*	−0.314	0.176†	−1.286	0.598*
SVO: prosself (C)	–	–	−0.704	0.420†	1.414	1.024
Gen. attitude vacc. (D)	–	–	–	–	0.043	0.009***
A * B	0.353	0.227	0.388	0.268	−0.261	0.953
A * C	–	–	0.320	0.622	−1.730	2.135
B * C	–	–	0.056	0.306	1.043	0.898
A * D	–	–	–	–	−0.065	0.014***
B * D	–	–	–	–	0.014	0.008†
C * D	–	–	–	–	−0.030	0.014*
A * B * C	–	–	−0.120	0.500	−2.094	1.860
A * B * D	–	–	–	–	0.008	0.012
A * C * D	–	–	–	–	0.030	0.028
B * C * D	–	–	–	–	−0.014	0.012
A * B * C * D	–	–	–	–	0.027	0.024
Observations (subjects/rounds/groups)			2795 (180/19/15)			
ML model fit: AIC/BIC	3202/3244		3207/3272		3179/3292	

* $p < .05$.** $p < .01$.*** $p < .001$.† $p < .1$.

Note that there are only 19 rounds considered in the analyses (rounds 2–20) because the first feedback was given *after* round 1.

vaccination game, i.e., the more positive the attitude, the more likely a decision is in favor of vaccination. In fact, a simple slope analysis reveals that whereas there is a positive and significant effect of vaccination attitude in the *Framing* treatment ($B = 0.042$, $SE = 0.009$, $p < .001$), there is a weaker and even negative effect in the *No-Framing* treatment ($B = -0.024$, $SE = 0.011$, $p = .034$). Moreover, a marginally significant interaction of RelVacc and the general vaccination attitude was found. A simple slope analysis reveals that players with a rather negative vaccination attitude chose vaccination strategically in response to the fraction of other players who chose vaccination (effect of RelVacc for players with a general vaccination attitude below the sample mean: $B = -0.605$, $SE = 0.256$, $p = .018$), whereas there was no effect for players with a rather positive vaccination attitude (effect of RelVacc for players with a general vaccination attitude above the sample mean: $B = -0.062$, $SE = 0.245$, $p = .800$). Finally, there is also a significant interaction effect of SVO and the general attitude towards vaccination, such that the attitude had a positive and significant impact on prosocials' ($B = 0.041$, $SE = 0.008$, $p < .001$) but no influence on proselves' decision in favor of vaccination ($B = 0.014$, $SE = 0.014$, $p = .312$).

4.3.3. Experienced costs and vaccination decisions

In a second set of analyses, the behavioral responses to experienced consequences of a participant's own decision in the previous round are investigated – vaccine-adverse events in the case of vaccination, and an infection with the disease in the case of non-vaccination (both labeled merely as point loss in the *No-Framing* treatment). Model 4 (see Table 2) predicts the vaccination decision by treatment and two dummy variables, representing the decision's consequences: one for the consequences of non-vaccination (0 = no point loss, 1 = point loss) and one for the consequences of vaccination (0 = no point loss, 1 = point loss; excluding the fixed costs due to vaccination).

Results of the regression suggest that point loss after non-vaccination (i.e., inaction) did not significantly increase vaccination uptake in the subsequent round. Thus, the experience of losing points due to illness did not significantly impact behavior in the next round. However, point loss after vaccination (i.e., action) significantly decreased the likelihood of vaccine uptake in the next round. Thus, if participants experienced adverse events after vaccination, they were more likely to change their behavior in the following round, even though the mean loss of points due to vaccine-adverse events was lower than after contracting the disease. This effect was independent of the treatment and mirrors the hypothesized omission bias.

Model 5 includes SVO as an additional predictor variable. A marginally significant effect of SVO provides further evidence that proselves were less likely to get vaccinated than prosocials. Finally, model 6 adds the general attitude towards vaccination. Again, a simple slope analysis indicates that prosocials' decisions in favor of vaccination were more affected by the general attitude towards vaccination ($B = 0.044$, $SE = 0.008$, $p < .001$) than proselves' decisions ($B = 0.014$, $SE = 0.014$, $p = .308$).

Table 2

Mixed effects models predicting decision in favor of vaccination by the experienced consequences in the previous round, experimental treatment (reference: *Framing* treatment), social value orientation (reference: prosocial) and general attitude towards vaccination.

Predictor	Model 4		Model 5		Model 6	
	B	SE	B	SE	B	SE
(Intercept)	0.186	0.206	0.447	0.248 [†]	−2.765	0.676 ^{***}
Treatment: No-Framing (A)	−0.242	0.296	−0.415	0.353	4.104	1.104 ^{**}
Point loss non-vaccination (B)	0.190	0.159	0.140	0.196	−0.050	0.565
Point loss vaccination (C)	−0.333	0.160 [†]	−0.433	0.188 [*]	−0.636	0.660
SVO: prosocial (D)	–	–	−0.832	0.430 [†]	1.526	1.046
Gen. attitude vacc. (E)	–	–	–	–	0.046	0.009 ^{**}
A * B	−0.073	0.229	0.004	0.272	0.691	0.909
A * C	0.192	0.234	0.364	0.276	0.315	1.030
A * D	–	–	0.489	0.640	−1.932	2.219
B * D	–	–	0.150	0.335	−0.112	0.942
C * D	–	–	0.346	0.356	0.064	1.071
A * E	–	–	–	–	−0.063	0.014 ^{***}
B * E	–	–	–	–	0.003	0.008
C * E	–	–	–	–	0.003	0.009
D * E	–	–	–	–	−0.033	0.014 [†]
A * B * D	–	–	−0.268	0.513	−0.338	1.950
A * C * D	–	–	−0.593	0.520	−1.217	1.896
A * B * E	–	–	–	–	−0.010	0.012
A * C * E	–	–	–	–	0.001	0.013
A * D * E	–	–	–	–	0.034	0.029
B * D * E	–	–	–	–	0.003	0.013
C * D * E	–	–	–	–	0.005	0.015
A * B * D * E	–	–	–	–	0.001	0.026
A * C * D * E	–	–	–	–	0.007	0.024
Observations (subjects/rounds/groups)			3420 (180/19/15)			
ML model fit: AIC/BIC	3879/3934		3886/3978		3876/4041	

^{*} $p < .05$.

^{**} $p < .01$.

^{***} $p < .001$.

[†] $p < .1$.

Note that there are only 19 rounds considered in the analyses (rounds 2–20) because the first feedback was given *after* round 1.

5. Discussion

The present paper integrates current developments in the epidemiological, economic, and psychological literature to analyze vaccination as a strategic interaction. We propose a novel behavioral game – the Interactive Vaccination (I-Vax) Game – to investigate individual vaccination behavior, taking into account its complex and dynamic incentive structure. This is the first attempt to study the behavioral consequences both of a vaccination's direct and indirect effects while considering the potential (negative) consequences of vaccination and non-vaccination. Hence, the I-Vax Game constitutes an appropriate game model to behaviorally simulate vaccination behavior of real-world vaccination decisions.

In line with the theoretical predictions, we find that participants react to the interactive structure of the I-Vax Game. They reduce their own likelihood of vaccination the more others got vaccinated in the previous round, especially when the game is framed as a vaccination decision. This indicates that, given that the incentive system is common knowledge, people make vaccination decisions strategically in order to maximize expected payoffs. We therefore conclude that non-vaccination can indeed be the result of selfish-rational behavior. Thus, vaccination can be seen as a social dilemma, not only from a theoretical, but also from a behavioral perspective. There are further psychological factors affecting individuals' vaccination decisions: As in other more simple social dilemmas (for a recent meta-analysis, see [Balliet et al., 2009](#)), prosocial actors are more likely to free-ride on others' vaccinations, whereas prosocials are prone to get vaccinated prosocially. Moreover, in the *Framing* treatment, particularly those individuals with a rather negative general attitude towards vaccination maximize their payoffs by free-riding on others' vaccination and thus benefitting from herd immunity. However, individuals are less "rational" than these findings might suggest, showing partially "biased" decision making. Only vaccine-adverse events resulted in a significant behavioral change, whereas an infection with the disease did not. This demonstrates the omission bias in vaccination decision making for the first time in a behavioral setting ([Spranca et al., 1991](#); [Ritov and Baron, 1990](#); [Kahneman and Tversky, 1982](#); [DeScioli et al., 2011](#); [Anderson, 2003](#)). Such an overestimation and fear of vaccine-adverse events may also explain why even after outbreaks vaccine uptake does not necessarily increase (e.g., after an outbreak of pertussis in Washington State; [Wolf et al., 2014](#)).

The present paper should be understood as an attempt to approximate the vaccination decisions' incentive structure in a laboratory game. The findings provide first evidence of the validity and robustness of the proposed game. For instance, when framing the game as a vaccination decision, individuals' general attitude towards real-life vaccination predicts vaccine uptake in a supposedly artificial game. In a similar vein, in the *Framing* treatment the amount of switching from vaccination to

non-vaccination and vice versa is affected by participants' vaccination attitude. Hence, the I-Vax Game opens up the possibility of studying vaccination behavior under controlled conditions by means of an incentivized behavioral game. Previous research, in contrast, was mainly based on simulations (e.g., [Bauch et al., 2013](#)) or survey studies (e.g. in the case of HPV vaccines, [Cox et al., 2010](#)), or did not satisfactorily map the epidemiological incentives and their psychological consequences into the game structure (mainly, by ignoring potential vaccine-adverse events, e.g., [Ibuka et al., 2014](#)).

5.1. Limitations

There are some things that could be improved in future research using the I-Vax Game. For instance, measuring vaccination attitudes directly after the game might lead to partially biased responses due to the experiences in the game. Although the general attitude towards vaccination appears to be a strong predictor of vaccination behavior even when controlling for experienced negative consequences due to the own decisions, we cannot fully exclude this possibility. Since measuring vaccination attitudes directly before an experiment might affect game behavior, we suggest to access this and further potentially interesting control variables that are related to real-world vaccinations (e.g., vaccinations in previous years) already some time before the actual experiment, e.g., when inviting participants to the experiment.

Additionally, although we provide feedback about other players' behavior after each round, it is reasonable to assume that players adapt behavior based on their own belief about other players' behavior in the following round. In fact, players' beliefs about other players future behavior (rather than other players' actual past behavior) could be an even better determinant of vaccination behavior and should be assessed in future studies. However, eliciting beliefs itself might have an effect on observed behavior (e.g. by increasing the salience; [Gächter and Renner, 2010](#)). For this reason we did not include belief measurements in this study to obtain baseline data regarding the effect of other people's behavior.

5.2. Further research

Future research could use this game-theoretical framework to detect mechanisms that may help to increase vaccination behavior and therefore decrease vaccine hesitancy. For instance, how and to what extent do social norms, rewards, or punishments influence vaccination behavior? Which institutional, ethical, normative, or monetary incentives can be used to increase vaccine uptake? Moreover, since the present paper provides results derived from fully informed participants, an important question regarding the influence of information arises: Do different kinds of feedback and incomplete information regarding the rules of the vaccination decision affect behavior (see e.g., [Helbing et al., 2014](#))? Because the acquisition of information regarding other persons' vaccination decisions requires some effort in real-life, it may also be interesting to raise costs for the acquisition of information. Furthermore, time preferences are likely to have an impact on preventive health behaviors like the vaccination decision, as it involves immediate costs and delayed consequences. For instance, [Chapman and Coups \(1999\)](#) found a relationship between vaccination acceptance and a measure of monetary time preferences. Hence, future research on the I-Vax Game could measure participants' time preferences to learn more about their relationship to vaccination decisions. Last but not least, vaccines are not 100% efficient, leaving some risk of infection even after vaccination. This is a particular problem of influenza vaccines, with a resulting efficacy of sometimes below 50% ([Kissling et al., 2014](#)). Future research may therefore investigate the behavioral consequences of inefficient vaccines using the I-Vax Game. The game paradigm may also be adapted to test the effect of vaccine scares, i.e., either rumors about vaccine safety ([Kata, 2012](#)), or the effect of very rare – but highly severe – adverse events. For instance, during the 2014 Ebola epidemics in Africa, the discussion has arisen about using non-licensed vaccines. Potential unexpected vaccine-adverse events may have delicate effects on vaccination behavior – driven by the demonstrated omission bias, but potentially also by a loss of trust in health organizations and vaccination programs. Adaptations of the I-Vax Game could serve to estimate the effect on vaccination behavior and further variants could include trust as a further determinant. Finally, considering the interactive incentive structure of health decisions is not only important to understand vaccination behavior, particularly vaccine hesitancy, but the general principle can also be expanded to other health behaviors. For example, in situations where it is not absolutely necessary, though possible, to take antibiotics, individuals also face a conflict between individual interest (e.g., fast recovery) and collective interest (e.g., antibiotic resistance due to overuse). Research in this domain will profit from incorporating the strategic-interactive aspects of the decision, which could be investigated using a similar game structure as proposed by the I-Vax Game.

5.3. Conclusion

We show that non-vaccination can be the result of selfish-rational considerations but also find evidence that additional factors increase (i.e., positive other-regarding preferences) or decrease (i.e., omission bias) vaccine uptake. We propose the I-Vax Game as a flexible experimental tool to pilot potential interventions in a controlled environment before their costly implementation in “the wild”. Providing an adaptive game paradigm, which is rooted in epidemiology, psychology and behavioral economics, we hope to stimulate the interdisciplinary exchange in this important research field to enlarge the understanding of vaccination behavior and vaccine hesitancy.

Appendix A. Supplementary Data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.jebo.2015.11.008>.

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